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Highly diastereoselective synthesis of quaternary α-trifluoromethyl α-amino acids by addition of benzylzinc reagents to chiral imines of trifluoropyruvate

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ABSTRACT

An effective and facile method for highly diastereoselective synthesis of quaternary α-trifluoromethyl α-amino acids was developed in good yields with high diastereoselectivity (dr >20:1) via $MgCl_2$-accelerated addition of benzylzinc reagents (Knochel type organozinc reagents) to (*R*)-phenylglycinol methyl ether based imines of trifluoropyruvate.

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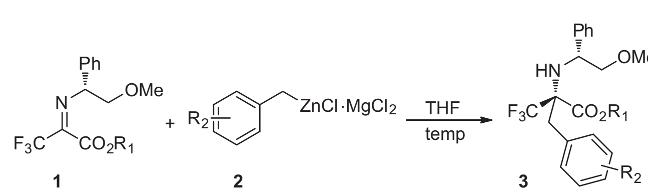
Fluorinated compounds have drawn extensive attention owing to their importance in materials and life sciences.[1] Among them, quaternary α-trifluoromethyl α-amino acids (α-Tfm-AAs) constitute a distinct class of fluorinated compounds. Because the unique properties of the CF_3 group,[2] such as high electronegativity, high lipophilicity, and steric hindrance that non-linearly increases with the number of fluorine atoms, can make such a special class of fluorinated amino acids greatly influence the bioavailability of peptides by improving enzymatic stability and enhancing in vivo absorption as well as drug permeability through certain body barriers.[3] In addition, due to the existence of quaternary chiral center in α-Tfm-AAs, once there is appropriate incorporation of α-Tfm-AAs into peptides, severe conformational restrictions are exerted on the peptide chains and thus can enhance the biological activity of the peptides by helping to pre-organize the optimum conformation for binding or by inhibiting metabolic degradation.[4] As a result, they can be used as peptidomimetics and therapeutic agents.[5] Furthermore, because of the abundance of ^{19}F in nature, CF_3 group can be used as a probe for kinetic and structural studies through ^{19}F NMR.[6] Hence, it is of great synthetic interest to develop an efficient method to synthesize these valuable fluorinated amino acids.

Although the most common method to non-racemic quaternary α-Tfm-AAs relies on the addition of organometallic reagents to chiral α-CF_3 ketoimine ester, the access of such a class of fluorinated amino acids in high diastereoselectivity using sp^3 hybridized organometallic reagents remains a problem. This is mainly because of the challenges in highly stereoselective construction of quaternary chiral center with a CF_3 group.[7] For example, the additions of chiral *p*-toluenesulfinyl α-CF_3 ketoimine ester with benzyl or alkyl magnesium halides only led to poor to moderate diastereoselectivity (dr 2.3–6.7).[8] In this regard, recently, we developed an efficient method for highly diastereoselective synthesis of quaternary α-allyl α-Tfm-AAs via indium mediated allylation of (*R*)-phenylglycinol methyl ether based imines of trifluoropyruvate **1**.[9] To continue our research interest, herein, we report an effective method for highly diastereoselective addition of benzylzinc reagents (Knochel type organozinc reagents)[10] to chiral α-CF_3 ketoimine ester **1**, which provides a facile access to α-Tfm-AAs (Scheme 1).

Considering that the low diastereoselectivity resulted from the addition of Grignard or organolithium reagents to chiral imines may arise from the active reactivities of these organometallic reagents, we envisioned that using a relatively 'inert' nucleophile to react with (*R*)-phenylglycinol methyl ether based imines of



Scheme 1. Diastereoselective synthesis of quaternary α-trifluoromethyl α-amino acids.

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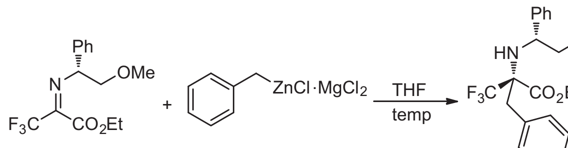
Table 1
Optimization of diastereoselective addition of benzyl zinc reagent **2a** to chiral α-CF_3 ketoimine ester **1a**[a]



1a **2a** **3a**

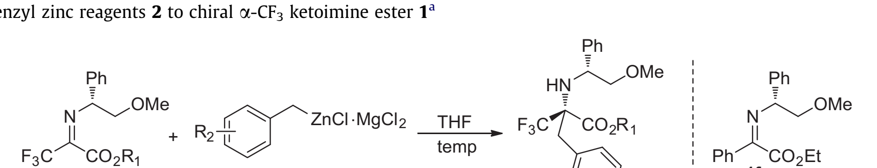
Entry	**2a** (equiv)	Temp (°C)	dr [b]	**3a**, yield (%)[c]
1	1.8	25	1.9:1	84
2	1.2	−10	>20:1	42
3	1.5	−10	>20:1	54
4	2.0	−10	>20:1	66
5	3.0	−10	>20:1	90
6	3.5	−10	>20:1	93
7	4.0	−10	>20:1	97
8	3.0	0	>20:1	90
9	3.5	0	>20:1	94 (93)
10[d]	3.5	0	4.3:1	9
11[e]	1.5	−10	>20:1	21
12[f]	1.5	−10	>20:1	47

[a] Unless otherwise noted, the reaction was carried out by using **1a** (0.5 mmol, 1 equiv), **2a** (1.2–4.0 equiv) in a 0.17 M THF solution (based on **1a**) for 8–9 h. For clarity complexed LiCl has been omitted.
[b] Determined by ^{19}F NMR before column chromatography.
[c] NMR yield determined by ^{19}F NMR, yield in parentheses was isolated yield.
[d] Benzyl zinc (3.5 equiv) was used without $MgCl_2$.
[e] Benzyl zinc (1.5 equiv) in conjunction with $MgCl_2$ (3.5 equiv) was used.
[f] Compound **2a** (1.5 equiv) in conjunction with $MgCl_2$ (2.0 equiv) was used.

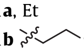
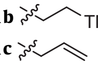
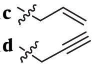
trifluoropyruvate **1** would help. In addition, the chelation of metal with N/O atoms in the compound **1** would lead to a rigid chelated transition state, as a result, a high diastereoselectivity would be observed. Recently, an important progress has been made by Knochel and coworkers,[10b] in which the addition of alkyl and benzyl zinc reagents to aldehydes or ketones could be realized by the presence of $MgCl_2$. Inspired by their research work, we began our studies of diastereoselective addition of chiral α-CF_3 ketoimine ester **1** by choosing Knochel type benzyl zinc reagent **2a** as nucleophile (Table 1). Although, initially a poor diastereoselectivity (dr 1.9:1) was provided when the reaction was carried out with **1** (1.0 equiv), benzyl zinc reagent **2a**[11] (1.8 equiv) in THF at room temperature (Table 1, entry 1), we were pleased to observe the formation of **3a** in high dr value (dr >20:1, determined by ^{19}F NMR) when the reaction was run at −10 °C with utilization of 1.2 equiv of **2a**, albeit in a low yield (42%) (Table 1, entry 2). Further optimization revealed that increasing the loading amount of **2a** benefited the reaction yield with no erosion of the diastereoselectivity (Table 1, entries 3–7), and high reaction yield could be obtained when 3.5–4.0 equiv of **2a** was used (Table 1, entries 6–7). Further increasing the reaction temperature to 0 °C with utilization of 3.5 equiv of **2a** still afforded **3a** in high diastereoselectivity with good yield (Table 1, entry 9). Interestingly, when **1a** was treated with only benzyl zinc reagent without $MgCl_2$, poor yield and diastereoselectivity were obtained (Table 1, entry 10). While, a higher yield and dr value were provided, when $MgCl_2$ was added (Table 1, entry 11). This finding demonstrated that $MgCl_2$ is essential for the reaction efficiency and diastereoselectivity, and may function as a Lewis acid.[10b] However, decreasing the loading amount of **2a** to 1.5 equiv in combination of 2.0 equiv of $MgCl_2$ only led to a low yield with a maintained diastereoselectivity (Table 1, entry 12), thus suggesting that an organozinc-magnesium complex is critical for the reaction. But the reason why excessive **2a** was required remains elusive and will be addressed in the future.

To ascertain the scope of this methodology, a variety of benzylzinc reagents **2** and chiral α-CF_3 ketoimine esters **1** were investigated (Table 2). Generally, excellent yields and high diastereoselectivities were obtained. For the electron-deficient benzylzinc reagents **2f** and **2g**, due to the formation of side products under the standard conditions, increasing loading amount of **2f** and **2g** to 4.0–5.0 equiv was found to benefit the reaction efficiency and diastereoselectivity (Table 2, entries 5–6). *para*- And *meta*-methyl benzylzinc reagents furnished the reaction smoothly with good yields and high dr values (Table 2, entries 1 and 3), but *ortho*-methyl substituted **2c** failed to give product **3c**, because of its poor reactivity (Table 2, entry 2). 2-Trimethylsilylethyl (TMSE), allyl and propargyl α-CF_3 ketoimine esters **1b–d** are also suitable substrates for the reactions, affording desired products in excellent

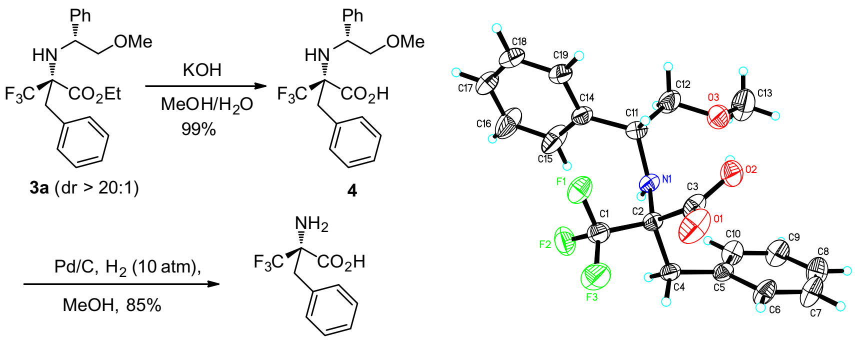
Table 2
Diastereoselective addition of benzyl zinc reagents **2** to chiral α-CF_3 ketoimine ester **1**[a]



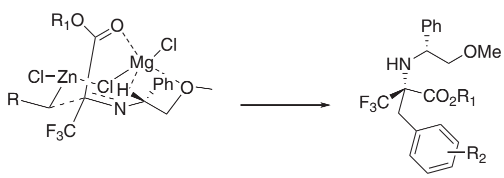
1 **2** **3**

Entry	**1** R_1	**2** R_2	Temp (°C)	Time (h)	dr[b]	**3**, yield[c] (%)
1	**1a**, Et	**2b**, 4-Me	0	8	>20:1	**3b**, 92
2	**1a**, Et	**2c**, 2-Me	0	24	—	**3c**, N.R.
3[d]	**1a**, Et	**2d**, 3-Me	−10	12	>20:1	**3d**, 95
4	**1a**, Et	**2e**, 4-OMe	0	8	>20:1	**3e**, 81
5[e]	**1a**, Et	**2f**, 4-F	−10	8	>20:1	**3f**, 91
6[d]	**1a**, Et	**2g**, 4-Cl	−10	12	>20:1	**3g**, 88
7[e]	**1b**  TMS	**2a**, H	0	8	>20:1	**3h**, 95
8	**1c** 	**2a**, H	0	12	>20:1	**3i**, 81
9	**1d** 	**2a**, H	0	12	>20:1	**3j**, 70
10	**1e**, Bn	**2a**, H	0	12	>20:1	**3k**, 95
11	**1f**	**2a**, H	25	12	—	**3l**, <10

[a] Unless otherwise noted, the reaction was carried out by using **1** (0.5 mmol, 1 equiv), **2** (3.5 equiv) in a 0.17 M THF solution (based on **1**). For clarity complexed LiCl has been omitted. N.R., no reaction.
[b] Determined by ^{19}F NMR before column chromatography.
[c] Isolated yield.
[d] Using 5.0 equiv of **2**, the reaction was run in a 0.08–0.10 M THF solution (based on **1**).
[e] Using 4.0 equiv of **2**, the reaction was run in a 0.15–0.17 M THF solution (based on **1**).



Scheme 2. Synthesis of (S)-α-Tfm-Phe **5** and X-ray crystal structure of compound **4**.



Scheme 3. The mechanistic proposal on stereocontrol.

yields with high dr values (Table 2, entries 7–10). It is noteworthy that these functional groups, TMSE, allyl, and propargyl, in α-Tfm-AAs can subsequently be applied for further functionalization, providing opportunities to access diverse α-Tfm-AAs derivatives. Unfortunately, when α-phenyl ketoimine ester **1f** was investigated, poor yield was obtained even at room temperature, thus indicating that an electron-deficient group, such as CF_3, in the ketoimine, is critical for the reaction efficiency (Table 2, entry 11).

The absolute configuration of quaternary chiral center in α-benzyl α-Tfm-AAs **3** was determined to be S by the X-ray crystallographic analysis of **4**.[12] As depicted in Scheme 2, compound **4** was synthesized from **3a** (dr >20:1) by treatment of KOH. Subsequently, hydrogenation of **4** in the presence of Pd/C afforded enantioenriched (S)-α-Tfm-Phe **5**[8] in 85% yield, demonstrating the synthetic utility of this method.

On the basis of our preliminarily results and Knochel's research work that $MgCl_2$ may function as Lewis acid and can chelate with carbonyl group to form a six-member ring chelation state,[10b] we proposed a highly restricted chair transition state for explanation of the high diastereoselectivity obtained in the reaction, in which Mg coordinates with C=N, MeO of imine, and carbonyl group of ester, which induces the benzylzinc reagents to attack the steric less hindered *Re* face to generate S configuration stereoselectively[13] (Scheme 3).

In conclusion, we have developed an effective and facile method for highly diastereoselective synthesis of quaternary α-Tfm-AAs in good yields with high diastereoselectivities by addition of Knochel type benzylzinc reagents to (R)-phenylglycinol methyl ether based imines of trifluoropyruvate. Although 3.0–5.0 equiv of benzyl zinc reagents were used, the low cost and ready availability of benzyl chlorides and the ease of preparation of Knochel type benzylzinc reagents make this protocol useful for application in peptides and related chemistry. Further studies to expand the substrates scope and their applications in synthesis and evaluation of bioactive compounds are currently in progress.

Acknowledgments

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Supplementary data

Supplementary data (detailed experimental procedures, and characterization data for new compounds) associated with this article can be found, in the online version, at doi:10.1016/j.tetlet.2011.07.006.

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11. Benzylzinc reagents were prepared according Knochel's procedure, see Ref. 10.
12. CCDC 821663 contains the supplementary crystallographic data for compound **4**. This data can be obtained free of charge from the Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.
13. The configuration of ketoimine ester **1a** is *E*. For the assignment of the configuration of **1a**, see: Chaume, G.; Van Severen, M.-C., Marinkovic, S.; Brigaud, T. *Org. Lett.* **2006**, *8*, 6123. However, in our previous work (see Ref. 9), a *Z* configuration of **1a** was drawn in the transition state. In this case, we thought that probably, there is an equilibrium between *E*-**1a** and *Z*-**1a** under the reaction conditions, and Z configuration is preferred in the transition state. However, an alternative transition state might be possible, see Supplementary data.